

GENERAL SPECIFICATION G4.63

Vessel Trays and Internals

CONTENTS

G4.63	Vessel Trays and Internals.....	1
G4.63.1	General.....	1
G4.63.2	Units of Measurement.....	1
G4.63.3	Codes and Standards	1
G4.63.4	Materials of Construction	1
G4.63.5	Bolting	2
G4.63.6	Gaskets and Packing.....	2
G4.63.7	Inspection.....	2
G4.63.8	Rejection	3
G4.63.9	Construction.....	3
G4.63.10	Structural Loads and Deflections	4
G4.63.11	Support Beams.....	5
G4.63.12	Support Welded to the Vessel (Tower Attachments).....	5
G4.63.13	Connections	7
G4.63.14	Corrosion Allowances and Minimum Metal Thicknesses	7
G4.63.15	Drainage and Leakage.....	7
G4.63.16	Fabrication	7

G4.63 VESSEL TRAYS AND INTERNALS

G4.63.1 General

- (a) This Specification defines mandatory requirements for the design, materials, fabrication, and assembly of removable vessel trays and related internals.
- (b) For the vessel Supplier, this Specification is intended only to provide a basis to include in the vessel quotation for the cost of furnishing and installing welded-in tower attachments and shop installation of trays/internals furnished by others.
- (c) In addition to the required quantity of tray hardware (bolts, nuts, washers, clamps, etc.) for installation, the supplementary materials required below shall be furnished:

Supplementary Material	
Parts	Additional Percent Beyond Design Quantities
Bolts, Nuts, Washers	10%
Tray Clamps	5%
Gaskets, Packing	20%

- (d) This specification shall be read in conjunction with the General Specification on Pressure Vessels.

G4.63.2 Units of Measurement

- (a) The S.I. units of measurement shall be used by the tray manufacturer for drawings and calculations.

G4.63.3 Codes and Standards

- (a) American Society of Mechanical Engineers (ASME)
ANSI/ASME Boiler and Pressure Vessel Code – Latest Edition, including all mandatory addenda in effect on the date of the Purchase Order:
Section VIII "Pressure Vessels" Division I
Section VIII "Pressure Vessels" Division II, Alternate Rules
Section IX "Welding, Brazing and Fusing Qualifications"
- (b) American Iron and Steel Institute
AISI "Cold-Formed Steel Design Manual," 2013 Edition

G4.63.4 Materials of Construction

- (a) All materials of construction shall be new and as specified. Material substitution is allowed only upon written acceptance from the S.O..

- (b) The nominal analysis of the tray material will be specified on the tray drawings or data sheets where available. Materials for tray components shall be selected from **Table G4.63-1** for the nominal analysis specified.
- (c) AISI bolt material may be proposed as an alternative to the ASTM bolt material specified in **Table G4.63-1**.

G4.63.5 Bolting

- (a) Bolting shall be marked to permit positive identification of the material.

G4.63.6 Gaskets and Packing

- (a) Gaskets and rope packing, when required, shall be of the non-asbestos tape type.

Table G4.63-1 - Tray Material Specifications			
Nominal Analysis	ASTM Specifications		
	Plate or Sheet	Bolts	Nuts
Carbon Steel	A-36, A-283 C, A-570 30 or 33, A-611 C	A-193 B6	A-194 6
12 Cr	A-240 405 or 410S 2D Finish A-176 405 or 410S 2D Finish	A-193 B6	A-194 6
304 S.S.	A-240 304 2D Finish A-167 304 2D Finish	A-193 B8 Class I	A-194 8
304L S.S.	A-240 304L 2D Finish A-167 304L 2D Finish	A-193 B8 Class I	A-194 8
316 S.S.	A-240 316L 2D Finish A-167 316L 2D Finish	A-193 B8M Class I	A-194 8M
316L S.S.	A-240 316L 2D Finish A-167 316L 2D Finish	A-193 B8M Class I	A-194 8M
321 S.S.	A-240 347 2D Finish A-167 321 2D Finish	A-193 B8T Class I	A-194 8T
347 S.S.	A-240 347 2D Finish A-167 347 2D Finish	A-193-B8C Class I	A-194 8C
Monel	B-127 Annealed	F-468 Alloy 400	F-467 Alloy 400

G4.63.7 Inspection

- (a) All materials, fabrications, and packaging shall be subject to inspection by the S.O. during all stages of manufacture.
- (b) All shop drawings shall be available to the S.O. at the time of inspection.
- (c) One tray of each diameter and each type for each vessel shall be assembled in the tray manufacturer's shop as a check for proper fit-up. When the test tray is a bubble cap type, it shall be leak tested as specified in the sub-section on Drainage and Leakage within this specification. All leak test certificates shall be submitted.

G4.63.8 Rejection

- (a) Materials containing defects, completed trays and parts of trays or vessel internals having improper fabrication or excessive repairs, or components that are not in complete compliance with the requirements of this specification and the General Specifications on Pressure Vessels will be rejected. Rectification or replacement with necessary removal from site of the defective item(s) shall be on a case by case basis and all method statements relating to these actions shall be subject to the approval of the S.O..
- (b) Discovery of conditions warranting rejection, after inspection and acceptance by the S.O., does not relieve the Contractor of responsibility to comply with this specification.

G4.63.9 Construction

- (a) General
 - (i) The nominal inside vessel diameter shall be as given on the tray drawings or data sheets where available. The nominal outside diameter of trays, pans, baffles, etc., shall be such that they overlap the support ring by a minimum of one-half of its width.
 - (ii) All parts shall be designed to pass through the vessel manholes. Exceptions to this requirement may be granted for large major beams and trusses, provided written approval of the S.O. is given prior to fabrication.
 - (iii) Bubble cap tray sections shall be designed to pass through the vessel manholes with the caps attached.
 - (iv) All like parts of tray assemblies shall be designed and made interchangeable.
 - (v) Trays shall be designed for assembly from above.
 - (vi) Packed tower internals such as gas injection plates (bed supports), liquid re-distributors, packing retaining plates, etc., shall be clamped to vessel support rings (unless otherwise specified) and designed for shop installation in the vessel with the vessel in a horizontal position.
 - (vii) Trays and draw-off pans, when assembled, shall have deck sections readily removable from both above and below manways.
 - (viii) Deck sections of bubble cap trays shall be removable with caps in place. When bubble caps are required to span more than one rise, individual riser caps shall be provided at removable tray sections.
 - (ix) Trays with centre down flows shall be provided with manways on both sides of the centre down comer. All removable sections for manways shall be located vertically in line and shall weigh no more than 20 kg or 50 pounds each. Manways shall be sized and located to permit easy access to all areas of the tray.
 - (x) All bolted or clamped joints in down comers of perforated or valve trays shall require gaskets; other joints in perforated or valve trays do not require gaskets.

- (b) All joints in bubble cap trays, including joints between the tray deck and support ring and down comers and down comer bolting bars, shall be gasketed. Bubble cap trays shall meet the leakage as specified.
- (c) All joints in seal pans and draw-off pans or sumps shall be gasketed.
- (d) Horizontal deck sections under down comers shall preferably be of one-piece construction suitable for passage through the vessel manhole. No obstructions to liquid flow shall be placed on deck sections under down comers.
- (e) The centre-to-centre spacing of bolting or clamping shall not exceed 150 mm.
- (f) Outlet weirs shall be adjustable plus or minus 13 mm from the specified weir height.
- (g) Inlet weirs shall be fixed.

G4.63.10 Structural Loads and Deflections

- (a) Trays, baffles, seal pans, and their supports shall be designed in accordance with the AISC Manual of Steel Construction or AISI Cold-Formed Steel Design Manual, whichever is applicable. Design shall be based on corroded thicknesses at design temperature. The yield strength and modulus of elasticity used for design shall be as specified in ASME Section II, Part D2 at design temperature. For the carbon steel materials listed in **Table G4.63-1** use the yield strengths listed for SA-285-C Material.
- (b) Structural design loads shall be as follows:
 - (i) Trays analytically designed by the tray manufacturer shall be designed for their dead load plus a uniform live load which is the greatest of 957 Pa or the weight of water on the tray at a depth of 50 mm over the maximum weir height. When specified on the data sheet or by the S.O. specifications, the trays shall be designed to accommodate uplift due to upset conditions.
 - (ii) Inlet areas (areas under the down comer) and seal pans shall be designed for their dead load plus a uniform live load which is the greatest of 3064 Pa or the weight of water at a depth of one-half the nominal tray spacing in the particular tower section.
 - (iii) Draw-off pans, accumulator trays or chimney trays shall be designed for their dead load plus a uniform live load equal to the maximum possible depth of the liquid (chimney height or overflow weir height) in the pan or tray.
 - (iv) Shed rows and disc and donut type baffles with no normal operating liquid level shall be designed for their dead load plus a uniform live load of 957 Pa on the projected horizontal area.
 - (v) Support beams and connections shall be designed for the total dead and live loads which they must carry. The effects of eccentrically applied loads producing torsion in the beams must be considered.
 - (vi) For maintenance purposes, trays, baffles, pans, etc. and their supports shall be designed for their dead load plus a concentrated load of 100 kg or 250 pounds at any location at a temperature of 38 °C or 100 °F.
 - (vii) The maximum deflection of a fully assembled tray from design loads shall not exceed the smaller of 1/900 of the vessel inside diameter or 4.8 mm.

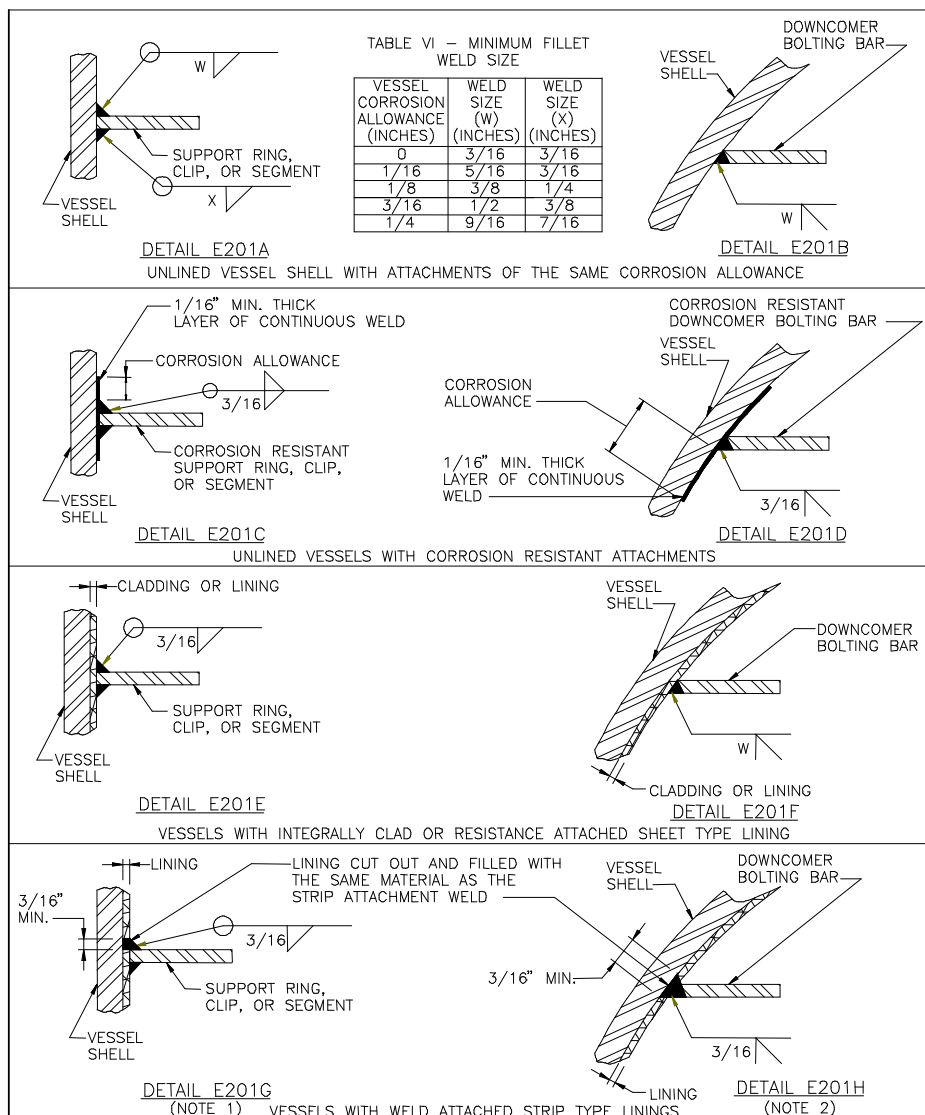
G4.63.11 Support Beams

- (a) The maximum depth of solid beam (not open truss type) tray and baffle supports perpendicular to the liquid flow on the tray below shall not exceed:
 - (i) 90 mm for tray spacing of 450 mm.
 - (ii) 25 % of the vertical tray spacing for trays spacing greater than 450 mm.
- (b) Support beams integral with deck plates are preferred.
- (c) Support beams or trusses may be cambered, subject to the S.O.'s approval. Camber shall not exceed the deflection due to dead load and shall be shown on the tray manufacturer's drawings.

G4.63.12 Support Welded to the Vessel (Tower Attachments)

- (a) The tray or packing supplier shall detail all tower attachments that will be needed to accommodate the tower internals.
- (b) The tower fabricator shall provide and install all welded tower attachments.
- (c) Supports to be welded to the vessel shall be in accordance with the details shown in **Table G4.63-2** (below) and **Figure 1** within this specification. The weld sizes shown are minimum values and shall be checked by the tray manufacturer for individual strength detailing the tower attachments:

Table G4.63-2 -Width of Support Rings	
Vessel Inside Diameter (1)	Support Ring Width (mm)
1 and under	32.5
Over 1 to 2	50
Over 2 to 4	62.5
Over 4 to 6	75
Over 6	75 minimum



Notes: The above welding details shall be used for support rings and attachments welded to vessel shells. Weld sizes are minimum values, and shall be checked for individual strength requirements. Alloy steel welds shall be used when either of the parts joined is alloy steel.

Numbered Notes:

1. Detail E201E may be used when the ring is located on, or is at maximum 25 mm above or below the strip attachment weld.
2. Detail E201F may be used when the bar spans, is located on, or is within 25mm of either side of the strip attachment weld.

Figure 1 - Welding Details

G4.63.13 Connections

- (a) Connections shall be designed in accordance with the AISC Manual of Steel Construction or the AISI Cold Formed Design Manual.
- (b) Joints between the tray and the support ring shall be "clamp type" with through bolting; threaded extruded-in clamps shall not be used. The clamped joint shall allow for thermal expansion from ambient to design temperatures.
- (c) Support member connections, and the top piece of the down comer-to-bolting-bar joints shall be through bolted joints.
- (d) Wedge type tray and bubble cap hold downs are not permissible.
- (e) The number of different bolt diameters shall be minimised for each vessel. When more than one bolt diameter is used, the bolts shall be paint marked for easy identification of bolt diameter.
- (f) The maximum centre-to-centre spacing of clamps or bolts shall be 150 mm, whether or not gaskets or packing are used.

G4.63.14 Corrosion Allowances and Minimum Metal Thicknesses

- (a) Corrosion allowance shall be added (on each surface) to the calculated thicknesses of all tower attachments and to the throat dimensions of welds attaching such parts to the vessel unless such parts are made of corrosion resistant material.
- (b) Corrosion allowance shall not be added to removable/replaceable parts (trays, baffles, etc.) not welded to the vessel.

G4.63.15 Drainage and Leakage

- (a) The number, size, and location of drain holes shall be specified on tray drawings or data sheets where available.
- (b) If drain holes are not located on the tray drawings or data sheets, 3/8 inch nominal diameter holes shall be provided as follows:
 - (i) One hole per every 1 m of length of inlet or outlet weir; and
 - (ii) One hole per every 1 m of length of seal pans, draw-off sumps, or other recessed areas.
- (c) For bubble cap trays, the assembled trays, with all manways bolted and drain holes plugged shall be sufficiently water tight to maintain a water level with a maximum leakage rate of 25 mm within a 3 minute period, unless otherwise specified. The initial water level shall be to the top of down flow weirs or risers, whichever is lower.

G4.63.16 Fabrication

- (a) Holes in perforated trays shall be punched from the top side of the trays.
- (b) Valves shall be provided with integral guide legs and lift stops and shall be dimpled or crimped to prevent the valve element from sitting flush on the tray deck. Flush-seating valves shall be furnished only when specified on the tray drawings or data sheets.

END OF SECTION